

Chapter 4: Image Restoration

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1. Image Enhancement vs. Image Restoration

Feature	Image Enhancement	Image Restoration
Nature	Subjective process	Objective process
Objective	Make image suitable for a specific application	Recover uncorrupted image from degradation
Mechanism	Manipulates visual features by preference	Applies inverse process with a priori knowledge
Applications	Photography, film archival, video encoding	Forensics, astronomy, medical imaging

2. Degradation & Restoration Model

Mathematical Formulation: $g(x,y) = H[f(x,y)] + n(x,y)$

Frequency Domain: $G(u,v) = H(u,v)F(u,v) + N(u,v)$

- **f(x,y):** Original uncorrupted input image
- **H:** Linear space-invariant degradation system operator
- **n(x,y):** Additive noise term
- **g(x,y):** Observed corrupted output image

3. Noise Models & Filtering

- **Noise Types:** Gaussian (electronic/sensor heat), Rayleigh (range imaging), Erlang/Gamma (laser), Salt & Pepper (electrical transient faults), Periodic (electromechanical interference).
- **Spatial Domain Filtering:** Adaptive Median Filters adapt behavior dynamically based on m x n local window statistics.
- **Frequency Domain Filtering:** Bandreject, Notch, and Optimum Notch filters isolate and eliminate periodic frequency spikes.

4. Degradation Function Estimation

- **Observation:** Inspecting degraded image patches alongside reconstructed unblurred references.
- **Experimentation:** Acquiring impulse responses under matched physical settings.
- **Mathematical Modeling:** Analytical modeling of environmental dynamics (e.g., motion blur, turbulence).